

2025 深中自招（回忆版）

1. $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4x}}} = \frac{3038}{2025}$, 则 $x =$ _____.

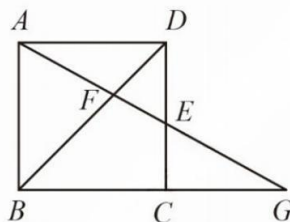
2. $x^2 + y^2 = 6xy, t = \frac{x+y}{x-y}$, 则 $60t^2 =$ _____.

3. $\frac{2025^3 - 1000^3 - 1025^3}{2025 \times 1000 \times 1025} =$ _____.

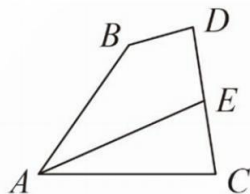
4. $\sqrt{x-1+2\sqrt{x-2}} - \sqrt{x-1-2\sqrt{x-2}} = 1$, 根的和为 t , $8t =$ _____.

5. $|x| + x + y = 10, |y| + y - x = 12$, $x + y$ 可能值和为 t , $20t =$ _____.

6. 在正方形 $ABCD$ 中, 在 BD 上取一点 F , 连接 AF 并延长交 BD 于点 F , CD 于点 E . 已知 $AF = 30$, $EF = 18$, 则 EG 的长度为 _____.

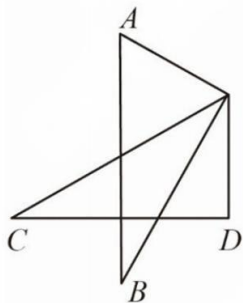


7. $AB = AE = AC = CD$, $BD = EC$, $BD \parallel EC$, 则 $\angle ABD$ 的度数为 _____.



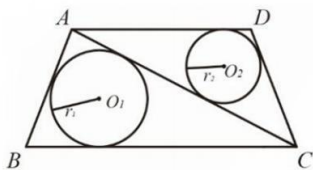
8. 已知 x 为实数且 $x < 1$, 设 $A = \frac{x^2 - 4x + 4}{1 - x}$, 则 A 的最小值为 _____.

9. 两个含 30° 角的直角三角板如图所示叠放在一起, $AB \perp CD$ 两三角板的面积均为 300, 则重叠部分面积为_____.



10. 已知 $\frac{x-2}{x^2+3}$ 的最大值为 M , 最小值为 n , $t = M - n$. 则 $36t^2 =$ _____.

11. 已知等腰梯形 $ABCD$ 中, $AB = CD = 10$, $BC = 21$, $\odot O_1, \odot O_2$ 分别切 $\triangle ABC$ 和 $\triangle ADC$ 则 $r_1 \cdot r_2 =$ _____.



12. $[x]$ 为不大于 x 的最大整数, 满足 $\sqrt{x^2 - [x]^2} = 3 - x$ 所有可能的 x 的值的和为 t , 则 $[t] =$ _____.

13. 已知 $|x^2 - ax + b| = 8$ 有三个不同的根, 三个根能组成一个直角三角形, 则 $|a - b|$ 所有可能值之和为_____.

14. 实数满足 $a_1 \leq a_2 \leq a_3 \leq \dots \leq a_{12}$, 任意三个数字之间无法组成锐角三角形. 则 $\frac{5a_{12}}{a_1}$ 的最小值是_____.

15. $B(n)$ 表示 $n = a_1 + 2a_2 + 2^2a_3 + \dots + 2^ka_k$ 的方法数, n 为自然数且 $n \geq 3, a_j = 0, 1, 2 (j < k), a_k = 1, 2$, 求 $B(2025) =$ _____.

2025 深中自招（回忆版）

1. $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4x}}} = \frac{3038}{2025}$, 则 $x = \underline{253}$.

$$1 + \frac{1}{1 + \frac{1}{4x}} = \frac{1}{\frac{\frac{3028}{2025} - 1}} = \frac{2025}{1013}$$

$$1 + \frac{1}{4x} = \frac{1}{\frac{2025}{1013} - 1} = \frac{1013}{1012}$$

$$4x = 1012, \Rightarrow x = 253.$$

2. $x^2 + y^2 = 6xy, t = \frac{x+y}{x-y}$, 则 $60t^2 = \underline{120}$.

$$(x+y)^2 = x^2 + y^2 + 2xy = 8xy$$

$$(x-y)^2 = x^2 + y^2 - 2xy = 4xy$$

$$t^2 = \left(\frac{x+y}{x-y}\right)^2 = 2.$$

$$\Rightarrow 60t^2 = 120.$$

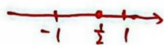
3. $\frac{2025^3 - 1000^3 - 1025^3}{2025 \times 1000 \times 1025} = \underline{3}$.

令 $1000 = a, 1025 = b$. 则

$$\text{原式} = \frac{(a+b)^3 - a^3 - b^3}{(a+b)ab} = \frac{3ab(a+b)}{(a+b)ab} = 3.$$

4. $\sqrt{x-1} + 2\sqrt{x-2} - \sqrt{x-1-2\sqrt{x-2}} = 1$, 根的和为 t , $8t = \underline{18}$.

$$|\sqrt{x-2} + 1| - |\sqrt{x-2} - 1| = 1$$



$$\Rightarrow \sqrt{x-2} = \frac{1}{2}$$

$$\Rightarrow x = \frac{9}{4}$$

$$\Rightarrow 8t = 18.$$

5. $|x| + x + y = 10, |y| + y - x = 12$, $x + y$ 可能值和为 t , $20t = \underline{160}$.

① $x \geq 0, y \geq 0$

$$\begin{cases} 2x + y = 10 \\ 2y - x = 12 \end{cases}$$

 $\Rightarrow \begin{cases} x = \frac{8}{5} \\ y = \frac{34}{5} \end{cases}$

② $x \geq 0, y < 0$

$$\begin{cases} 2x + y = 10 \\ -x = 12 \end{cases}$$

 $\Rightarrow \begin{cases} x = -12 \\ y = 34 \end{cases}$ 舍

③ $x < 0, y \geq 0$

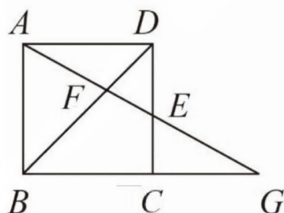
$$\begin{cases} y = 10 \\ 2y - x = 12 \end{cases}$$

 $\Rightarrow \begin{cases} y = 10 \\ x = 8 \end{cases}$ 舍

④ $x < 0, y < 0$

$$\begin{cases} y = 10 \\ -x = 12 \end{cases}$$
 舍
 $\therefore 20t = 20 \times (\frac{8}{5} + \frac{34}{5}) = 160$.

6. 在正方形 $ABCD$ 中, 在 BD 上取一点 F , 连接 AF 并延长交 BD 于点 F , CD 于点 E . 已知 $AF = 30$, $EF = 18$, 则 EG 的长度为 32.



令 $DE = 18t$, 则 $AB = 30t$, $CE = 12t$.

$\because AD \parallel CG$

$\therefore EG = \frac{CE}{DE} \cdot AE = \frac{12t}{18t} \times 48 = 32$.

7. $AB = AE = AC = CD$, $BD = EC$, $BD \parallel EC$, 则 $\angle ABD$ 的度数为 140°.

过点 A 作 CD 的平行线.

过点 C 作 BD 的平行线, 延长 DB 连接 BC

则 $\square MAED$, $\square ANCE$

$ME = AE = NC = CD$

$\therefore \square MDCN$ 为菱形

$\therefore BD = CE = AN$

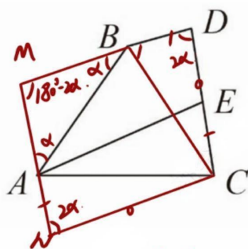
$\therefore MB = MA$

设 $\angle MBA = \angle MAB = \alpha$

证 $\triangle BDC \cong \triangle ANC$ (SAS)

$\therefore BC = AC$

$\therefore BC = AC = AB$



$\therefore \triangle ABC$ 为等边三角形

$\therefore CD = AC$

$\therefore CD = BC$

$\therefore \angle BDC = \angle D = 2\alpha$

$\therefore \angle MBA + \angle ABC + \angle DBC = 180^\circ$

即 $\alpha + 60^\circ + 2\alpha = 180^\circ$

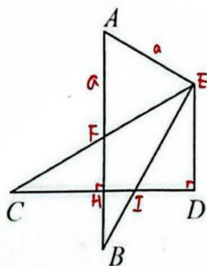
$\alpha = 40^\circ$

$\therefore \angle ABD = 180^\circ - \angle MBA = 140^\circ$

8. 已知 x 为实数且 $x < 1$, 设 $A = \frac{x^2 - 4x + 4}{1 - x}$, 则 A 的最小值为 4.

$$\begin{aligned} A &= -x + 3 + \frac{1}{-x+1} \\ &= (-x+1) + \frac{1}{-x+1} + 2 \\ &\geq 2\sqrt{(-x+1) \cdot \frac{1}{-x+1}} + 2 = 4 \\ &\text{当且仅当 } -x+1 = \frac{1}{-x+1} \text{ 时, 即 } x=0 \text{ 时, " " 成立.} \end{aligned}$$

9. 两个含 30° 角的直角三角板如图所示叠放在一起, $AB \perp CD$ 两三角板的面积均为 300, 则重叠部分面积为 125.



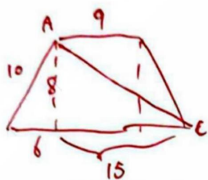
易证: $\triangle AFE$ 为等边三角形.

$$\begin{aligned} \text{令 } AE = AF = EF = a, \text{ 则 } CF = CE - EF = a. \therefore FH = \frac{a}{2}. \therefore BH = \frac{a}{2}. \therefore S_{\triangle BHI} &= \left(\frac{\frac{a}{2}}{\sqrt{3}a}\right)^2 \cdot S_{\triangle ABE} = \frac{1}{12} S_{\triangle ABE} \\ \therefore S_{\text{重}} = S_{\triangle EFB} - S_{\triangle HBI} &= \frac{1}{2} S_{\triangle ABE} - \frac{1}{12} S_{\triangle ABE} = \frac{5}{12} \times 300 = 125. \end{aligned}$$

10. 已知 $\frac{x-2}{x^2+3}$ 的最大值为 M , 最小值为 n , $t = M - n$. 则 $36t^2 =$ 28.

$$\begin{aligned} \text{令 } y &= \frac{x-2}{x^2+3}, \text{ 则} \\ yx^2 - x + 3y + 2 &= 0 \\ \therefore \Delta &= 1 - 4y(3y+2) \\ &= -[12(y+\frac{1}{3})^2 - \frac{7}{3}] \geq 0 \\ \therefore (y+\frac{1}{3})^2 &\leq \frac{7}{36} \\ \therefore -\frac{\sqrt{7}}{6} - \frac{1}{3} &\leq y \leq \frac{\sqrt{7}}{6} - \frac{1}{3} \\ \therefore M - N &= \frac{\sqrt{7}}{3} \\ \therefore 36t^2 &= 36 \times \left(\frac{\sqrt{7}}{3}\right)^2 = 28. \end{aligned}$$

11. 已知等腰梯形 $ABCD$ 中, $AB=CD=10$, $BC=21$, $\odot O_1, \odot O_2$ 分别切 $\triangle ABC$ 和 $\triangle ADC$ 则 $r_1 \cdot r_2$ = 7 .

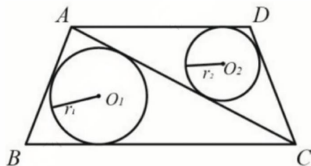


$$AC = \sqrt{8^2 + 15^2} = 17.$$

$$r_1 = \frac{S}{P} = \frac{\frac{1}{2} \times 21 \times 8}{\frac{1}{2} \times (10+21+17)} = \frac{7}{2}$$

$$r_2 = \frac{S}{P} = \frac{\frac{1}{2} \times 9 \times 8}{\frac{1}{2} \times (9+10+17)} = 2$$

$$\therefore r_1 r_2 = 7.$$



12. $[x]$ 为不大于 x 的最大整数, 满足 $\sqrt{x^2 - [x]^2} = 3 - x$ 所有可能的 x 的值的和为 t , 则 $[t]$ = 6 .

$$\because 3-x \geq 0$$

$$\therefore x \leq 3$$

$$x^2 - [x]^2 = x^2 - 6x + 9$$

$$\Rightarrow [x]^2 = 6x - 9$$

$$又 0 \leq 6x - 9 \leq 9$$

$$\therefore [x]^2 = 0 \text{ 或 } 1 \text{ 或 } 4 \text{ 或 } 9$$

$$\textcircled{1} [x] = 0, \quad 6x - 9 = 0, \quad x = \frac{3}{2} \text{ 成立.}$$

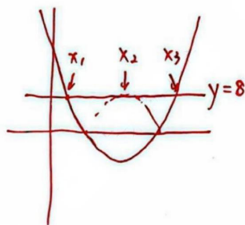
$$\textcircled{2} [x] = 1, \quad 6x - 9 = 1, \quad x = \frac{5}{3} \text{ 成立.}$$

$$\textcircled{3} [x] = 2, \quad 6x - 9 = 4, \quad x = \frac{13}{6} \text{ 成立.}$$

$$\textcircled{4} [x] = 3, \quad 6x - 9 = 9, \quad x = 3 \text{ 成立.}$$

$$[t] = \left[\frac{3}{2} + \frac{5}{3} + \frac{13}{6} + 3 \right] = 6$$

13. 已知 $|x^2 - ax + b| = 8$ 有三个不同的根, 三个根能组成一个直角三角形, 则 $|a-b|$ 所有可能值之和为 216 .



如图, $x^2 - ax + b$ 的最小值为 -8 .

$$\therefore \frac{4b - a^2}{4} = -8.$$

$$\therefore a^2 - 4b = 32 \quad \textcircled{1}$$

$$\text{令 } x_1 < x_2 < x_3, \text{ 则 } x_3 = \frac{a}{2}.$$

$$x_3^2 - x_1^2 = x_2^2$$

$$\therefore (x_3 + x_1)(x_3 - x_1) = x_2^2$$

$$\therefore a \cdot \sqrt{a^2 - 4(b-8)} = \left(\frac{a}{2}\right)^2$$

$$\therefore a^2 - 4a \sqrt{a^2 - 4b + 32} = 0$$

$$\text{由 } \textcircled{1} \text{ 得 } a^2 - 4a \sqrt{64} = 0 \text{ 即 } a(a-12) = 0$$

$$1^\circ a = 0, \quad x_2 = 0, \text{ 矛盾}$$

$$2^\circ a = 12, \quad b = \frac{a^2 - 32}{4} = 248.$$

$$\therefore |a-b| = 216.$$

14. 实数满足 $a_1 \leq a_2 \leq a_3 \leq \dots \leq a_{12}$, 任意三个数字之间无法组成锐角三角形. 则 $\frac{5a_{12}}{a_1}$ 的最小值是 60.

$$1, 1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{8}, \sqrt{13}, \sqrt{21}, \sqrt{34}, \sqrt{55}, \sqrt{89}, \sqrt{144}$$

$$\frac{5 \times 12}{1} = 60.$$

15. $B(n)$ 表示 $n = a_1 + 2a_2 + 2^2a_3 + \dots + 2^k a_k$ 的方法数, n 为自然数且 $n \geq 3, a_j = 0, 1, 2 (j < k)$, $a_k = 1, 2$, 求 $B(2025) =$ 33.

$$2025 = \underbrace{(11111101001)}_m \underbrace{1}_{n} \cdot 2$$

$$1 = \begin{cases} 1001 \\ 0201 \\ 0121 \end{cases} \quad 3 \times 7.$$

$$m = \begin{cases} 1111110 \\ 1111102 \\ 1111022 \\ 1110222 \\ 1102222 \\ 1022222 \\ 2222222 \end{cases}$$

当 $n=1001$ 时, 无.

当 $n=0201$ 时, 68种

当 $n=0121$ 时, 68种

$$\therefore 3 \times 7 + 6 + 6 = 33$$